

PROJECT TOPIC: BIODIVERSITY, FOOD CHAINS AND FOOD WEBS

Topic Description: Ecological Study of a Local Natural Habitat to Map Trophic Levels, Food Chains, and Complex Food Webs

In every natural ecosystem on Earth, living organisms are inextricably bound together through intricate networks of trophic feeding relationships. The fundamental law of ecology dictates that no single organism exists in isolation; energy captured from sunlight by green plants flows continuously through successive feeding levels—from primary producers to herbivores, carnivores, apex predators, and decomposers.

This project presents a structured ecological investigation of a **Local Natural Area** (surveying a representative semi-natural grassland and scrub forest habitat in Maharashtra) to observe biodiversity, document trophic levels, and reconstruct authentic **Food Chains and Food Webs** functioning within the community.

Understanding the difference between a simplistic linear food chain and a resilient, multi-linked food web is critical for environmental science students. By mapping multiple feeding pathways, the study illustrates how higher biodiversity provides ecological stability: when one prey species fluctuates, predators switch to alternate food sources, maintaining dynamic ecological balance.

2. Importance of the topic (विषयाचे महत्त्व)

Studying food chains and webs provides essential insights into ecosystem stability, bioaccumulation, and conservation biology:

Key Ecological Significance :-

- **Ecosystem Homeostasis & Resilience:** Food webs illustrate that ecological stability depends on complexity. Monocultures with single-path food chains collapse easily when a single pest attacks, whereas diverse natural food webs absorb shocks and self-stabilize.
- **Unidirectional Energy Flow & 10% Rule:** Demonstrates Lindeman's fundamental 10% ecological efficiency law: roughly 90% of energy is dissipated as metabolic heat at each trophic transfer, explaining why top predators (tigers, eagles) are naturally rare and require vast territories.
- **Biological Magnification of Toxic Pollutants:** Visualizing food webs is critical for tracking how persistent non-biodegradable pesticides (like DDT and heavy metals) biomagnify exponentially up the trophic ladder, causing reproductive failure in apex raptors.
- **The Crucial Role of Decomposers:** Highlights the often-ignored detritus food chain—soil fungi, earthworms, and bacteria—which recycles locked organic nutrients back into the soil for plant uptake.

The study proves that protecting top predators requires conserving every underlying link in the trophic web.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

The field project was designed to fulfill structured ecological and trophic mapping objectives:

Structured Project Objectives :-

- • **Habitat Demarcation & Survey:** To delineate a 1-hectare representative grassland and deciduous scrub forest area and record abiotic features (sunlight, soil type, moisture).
- • **Trophic Level Categorization:** To observe, identify, and categorize resident organisms into Producers (T1), Primary Consumers (T2), Secondary Consumers (T3), Tertiary Consumers (T4), and Decomposers (T5).
- • **Direct Food Chain Construction:** To construct three authentic, verified grazing food chains and one detritus food chain based on direct field observations.
- • **Complex Food Web Synthesis:** To map intersecting feeding interactions and synthesize a comprehensive ecological food web diagram showing alternative energy pathways.
- • **Keystone Species Assessment:** To evaluate the ecological consequences on the entire food web if a keystone producer or top predator is removed.

The scope encompasses direct morning field surveys, scat and pellet observations, and trophic literature verification.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A standardized observational ecological survey protocol was deployed to map trophic interactions:

Methodological Protocol Executed :-

- **Phase 1 – Field Site Selection:** Selected a 1-hectare mosaic of native grassland and thorny scrub forest at the base of a local hill slope in Maharashtra.
- **Phase 2 – Direct Biological Observation:** Conducted early morning (6:30 AM to 9:30 AM) and evening (4:30 PM to 6:30 PM) observation transects over four field visits using 8x42 binoculars.
- **Phase 3 – Trophic Feeding Evidence Recording:** Documented direct feeding acts (caterpillars on leaves, birds catching insects, raptor feeding perches) and indirect evidence (chewed leaves, regurgitated owl pellets, insect galls).
- **Phase 4 – Taxonomic Classification:** Classified all observed plants, insects, reptiles, birds, and mammals using field keys and Balbharati EVS references.
- **Phase 5 – Network Construction:** Interlinked verified predator-prey relationships into a structured trophic matrix and food web chart.

All feeding relationships were verified against peer-reviewed zoological dietary literature.

5. Observations (निरीक्षणे)

Systematic observations across the 1-hectare scrub-grassland study site documented diverse organisms across five trophic levels:

5.1 Trophic Classification of Observed Organisms

Trophic Level	Representative Observed Species	Functional Ecological Role in Ecosystem
T1: Producers	Dhaman grass (<i>Cenchrus</i>), Spear grass, Ber shrub (<i>Ziziphus</i>), Khair tree (<i>Acacia catechu</i>)	Autotrophs; convert solar energy into biochemical energy through photosynthesis; form the biomass base.
T2: Primary Consumers (Herbivores)	Grasshopper, Plain Tiger caterpillar, Indian Hare, Field Mouse, Desert Locust, Cattle	Herbivores; consume plant biomass, converting vegetative cellulose into animal protein for predators.
T3: Secondary Consumers (Carnivores)	Garden Lizard (<i>Calotes</i>), Praying Mantis, Ashy Prinia (insectivorous bird), Green Bee-eater, Toad	Primary carnivores; actively hunt herbivorous insects, maintaining insect populations below pest-bloom levels.
T4: Tertiary Consumers (Apex Predators)	Black Kite (<i>Milvus migrans</i>), Common Kestrel, Indian Cobra (<i>Naja naja</i>), Russell's Viper, Barn Owl	Top predators; regulate populations of rodents, lizards, and small birds, preventing overgrazing of vegetation.
T5: Decomposers & Detritivores	Termites, Earthworms, Dung Beetles, Soil Fungi (<i>Aspergillus</i>), Actinomycetes bacteria	Decomposers; break down dead animal carcasses, fallen leaves, and dung, recycling vital nitrogen and phosphorus.

5.2 Verified Linear Food Chains Recorded

• Chain 1 (Grassland): Grass → Grasshopper → Garden Lizard → Indian Cobra →

6. Analysis of data (माहितीचे विश्लेषण)

Ecological analysis of the documented food web reveals key principles of energy flow and ecological vulnerability:

Core Analytical Inferences :-

- **Food Web Redundancy Equals Stability:** The constructed food web demonstrates that generalist predators enjoy dietary flexibility. The Indian Cobra feeds on field mice, toads, and garden lizards; if disease depresses lizard populations, the cobra shifts predation pressure to rodents, preserving overall community stability.
- **Top-Down Trophic Cascade Vulnerability:** Indiscriminate killing of snakes and owls by local residents leads to an immediate population explosion of herbivorous field rodents, which inflict severe agricultural damage on nearby wheat and vegetable crops.
- **Energy Pyramid Constraints:** The pyramid of biomass shows a massive base of grasses (T1), narrowing sharply to fewer herbivores (T2), fewer lizards (T3), and only 2 nesting pairs of apex Black Kites (T4), directly validating Lindeman's 10% thermodynamic energy transfer limit.
- **Pesticide Disruption of Food Chains:** Chemical weedicide spraying along road verges kills wild grasses and pollinators simultaneously, causing insectivorous birds (Prinias, Bee-eaters) to abandon the territory due to sudden starvation.

The analysis proves that every link in the food web performs an indispensable regulatory function.

7. Results and Conclusions (निष्कर्ष)

Conclusion: The field study confirms that local semi-natural grasslands function as vibrant, delicately balanced ecological communities. Human conservation must protect entire food webs rather than isolated species, ensuring that natural energy flows and decomposition cycles continue uninterrupted.

Action Checklist for Food Web Stewardship :-

- ❑ Protect non-venomous and venomous snakes from senseless killing; recognize their vital role as natural rodent controllers.
- ❑ Avoid using chemical rodenticides, which cause secondary poisoning of predatory owls and hawks.
- ❑ Retain wild patches of native grasses and thorny shrubs on campus to support insect and bird food chains.
- ❑ Stop broad-spectrum pesticide spraying that wipes out essential herbivorous and pollinating insects.
- ❑ Educate farming families on the economic benefits of natural pest predators (toads, mantises, kites).

8. Bibliography / References (संदर्भ सूची)

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